Trainings

General Information

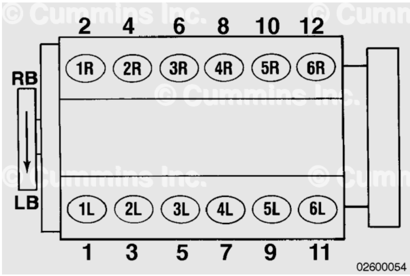
Valves **must** be adjusted correctly for the engine to operate efficiently. Valve adjustment **must** be performed using the values listed in this procedure.

Engine firing order: 1R-1L-5R-5L-3R-3L-6R-6L-2R-2L-4R-4L

Cylinders are numbered from the front gear cover end of the engine.

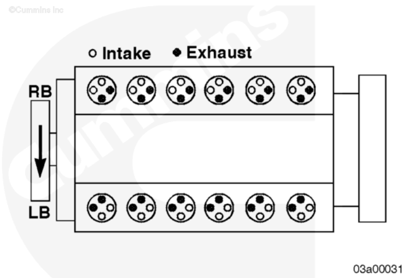
To determine the right and left banks on the engine, stand at the rear of the engine and face the front.

Two crankshaft revolutions are required to adjust all of the valves.



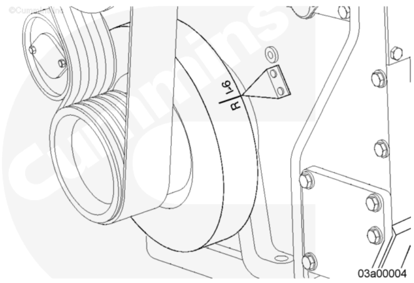
Each cylinder has two rocker levers. On the left bank, the lever nearest to the rear of the engine is the intake lever. On the right bank, the exhaust valve is nearest to the rear.

One pair of valves are adjusted at each index mark before rotating the engine to the next index mark.

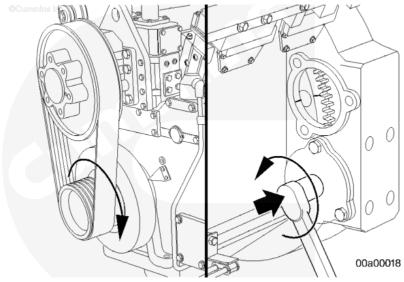


Adjust

Valve adjustment marks are on the vibration damper. The marks **must** be aligned with the pointer.

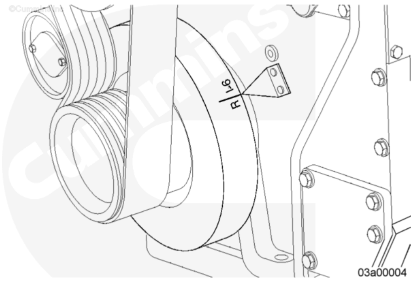


Rotate the engine **clockwise**, as the arrow on the pulley indicates, with a barring tool.



Check that the number 1 right-bank valves are ready to be adjusted, by seeing that both rocker levers on the number 1 right bank are loose and the rocker levers on the number 6 left bank are uneven.

If **not**, rotate the engine 180 degrees.



QST30 Valve Adjustment Chart		
Vibration Damper Mark	**Valves Closed or Push Rod Protrusion Even on Cylinder Number**	**Adjust Valves on Cylinder Number**
R1-6	1RB	1RB
L1-6	1LB	1LB
R2-5	5RB	5RB
L2-5	5LB	5LB
R3-4	3RB	3RB
L3-4	3LB	3LB
R1-6	6RB	6RB
L1-6	6LB	6LB
R2-5	2RB	2RB
L2-5	2LB	2LB
R3-4	4RB	4RB
L3-4	4LB	4LB

The rocker levers will be the same height on the cylinder ready for valve adjustment.

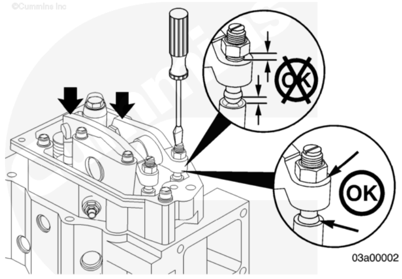
The crossheads and valves are ready to be adjusted on the cylinder that has all the valves closed.

Check the two cylinders shown by a mark on the vibration damper.

If the rocker lever assemblies have been removed, use this step to determine the cylinder to set.

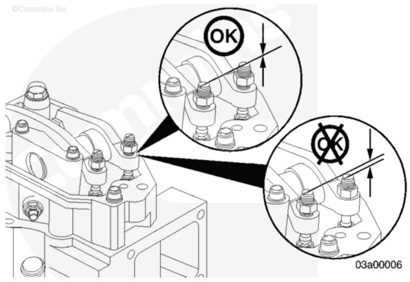
All adjusting screws **must** be loose on all cylinders and the push rods **must** remain in alignment.

Hold both rocker levers against the crossheads. Turn the adjusting screws until they touch the push rods. Turn the locknuts until they touch the levers. Tighten the adjusting screws an additional 20 degrees.

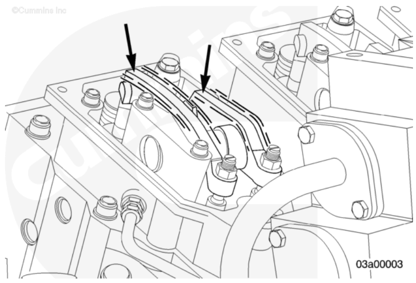


The cylinder with the adjusting screws that are nearly the same height is ready for valve adjustment. The second cylinder that is **not** ready for adjustment will have the adjusting screw for the intake valves more than five threads above the exhaust screw.

The rocker levers will be close to the same height above the top of the rocker lever housing on the cylinder ready for valve adjustment.

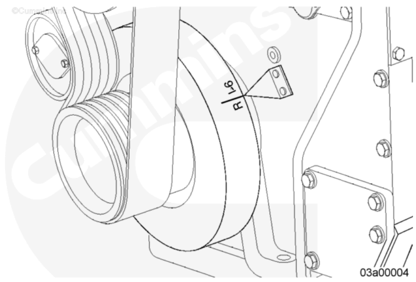


If the rocker levers have **not** been removed and installed again, wiggle the valve rocker levers on the two cylinders in question. The crossheads and valves on the cylinder where both levers feel loose are ready to adjust.



⚠ CAUTION ⚠

Use the correct firing order for the engine. Failure to do so can cause engine damage.



Adjustment can begin on any valve set mark. In the example, assume the R1-6 mark is aligned and the adjusting screw height for the valve on cylinder number 1 right bank is the same, indicating the valve is closed and ready to adjust.

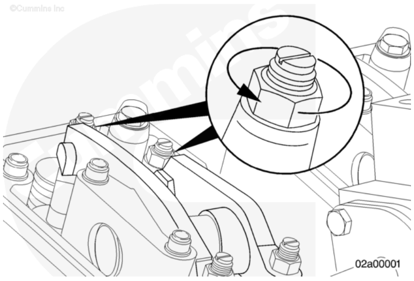
After identifying which cylinder is ready to be adjusted, follow the engine firing order for subsequent adjustments.

The firing order for the engine is 1R-1L-5R-5L-3R-3L-6R-6L-2R-2L-4R-4L.

Crosshead adjustment **must always** be made before attempting to adjust the valves.

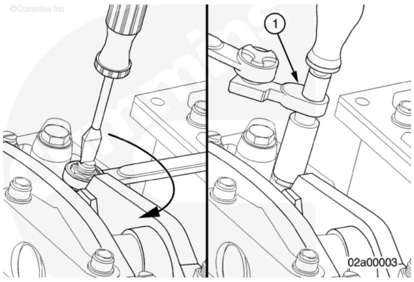
Adjust the crossheads on the cylinder that has both valves closed.

Loosen the crosshead adjusting screw locknuts on the intake and exhaust valve crossheads.



Adjust the crosshead.

- Loosen the locknut and take the tension off the adjustment screw.
- Holding the crosshead in place, tighten the adjustment screw until contacts the valve stem.
- Tighten an additional 20 degrees to remove any residual oil in the cup of the crosshead and ensure proper valve lash.
- Tighten the locknut.



The adjusting screw **must not** turn when the locknut is tightened.

Tighten the locknut using torque wrench service tool, Part Number ST-699, with adapter (1).

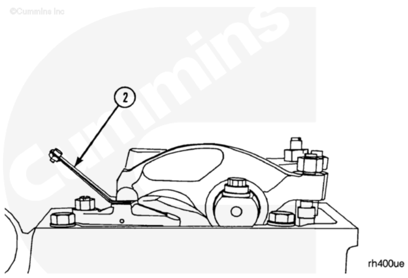
Torque Value: 45 n•m [33 ft-lb]

Tighten the locknut using torque wrench service tool, Part Number ST-699, without adapter.

Torque Value: 60 n•m [44 ft-lb]

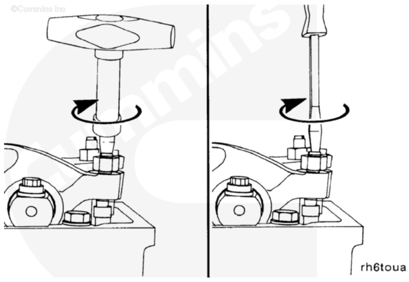
Select a feeler gauge for the correct valve lash specification.

Insert the gauge (2) between the rocker lever and the crosshead.



Intake and Exhaust Valve Adjustment		
Valve	**mm**	**[in]**
Exhaust	0.80	0.032
Intake	0.43	0.017

Two different methods for establishing valve lash clearance are described below. The two methods are torque wrench method or feel method; either method can be used; however, the torque wrench method has proven to be the most consistent.

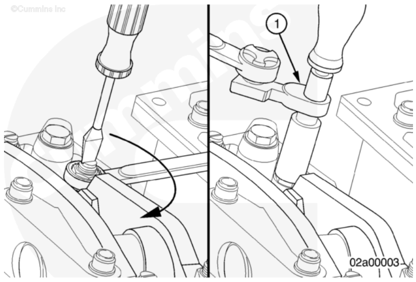


Use a service tool inch pound torque wrench, Part Number 3376592.

Torque Value: 0.68 n•m [6 in-lb]

For the feel method, use a screwdriver to turn the adjusting screw **only** until the lever touches the feeler gauge.

The adjusting screw **must not** turn when the locknut is tightened.



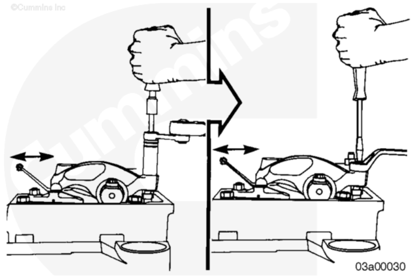
Tighten the locknut using torque wrench service tool, Part Number ST-699, with adapter (1).

Torque Value: 45 n•m [33 ft-lb]

Tighten the locknut using torque wrench service tool, Part Number ST-699, without adapter.

Torque Value: 60 n•m [44 ft-lb]

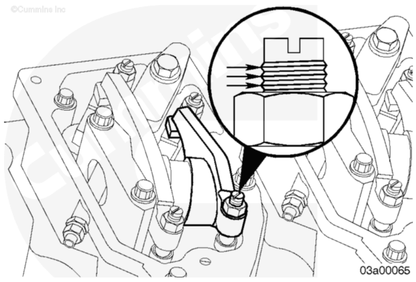
The feeler gauge **must** slide backward and forward with **only** a slight drag.



Attempt to insert a feeler gauge that is 0.03 mm [0.001 in] thicker. The valve lash is **not** correct when the thicker gauge will fit.

Repeat the adjustment process until the clearance is correct on both the intake and the exhaust valves on the cylinder being adjusted.

After the overhead has been completely set, check the adjusting screw protrusion. Approximately three threads **must** be exposed above the locknut.



If **not**, look for:

- Overhead set improperly
- Rocker housing gasket missing
- Push rod **not** in socket
- Cam follower socket **not** seated properly
- Nut thickness **not** standard
- Crosshead off valve stem.

Last Modified: 13-Oct-2022